26900163 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECFS302 - EMBEDDED SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Describe the purpose of emulation. [U] / [CO1]
2. Explain what a DSP processor is. [U] / [CO1]
3. Classify embedded systems based on their complexity. [U] / [CO1]
4. Describe one modeling technique used in embedded systems. [U] / [CO1]
5. Identify any two applications of UART. [U] / [CO2]
6. Identify the functions of the TxD and RxD pins in RS-232 communication. [U] / [CO2]
7. Identify the two signal lines used in a typical CAN bus and state their functions. [U] / [CO2]
8. Explain any two characteristics of an RTOS. [U] / [CO3]
9. Explain priority-based task scheduling. [U] / [CO3]
10. Explain source calls in interrupt handling. [An] / [CO3]

11. Examine the importance of device drivers in device management. [An] / [CO3]

12. A user changes an alarm from 6:00 AM to 7:00 AM. Show what changes in the alarm clock. [A] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain the importance of simulation tools in embedded system design. [U] / [CO1]

14. Explain the essential characteristics of embedded systems with examples. [U] / [CO1]

15. Comprehend the relationship between Wi-Fi frequency, range, speed, and interference, and explain how these factors influence the user's experience. [U] / [CO2]

16. Describe how low power consumption, low cost, and reliable communication make Zigbee suitable for IoT applications. [U] / [CO2]

17. Explain the major services provided by an operating system. [U] / [CO3]

18. Discuss the effect of context-switching overhead on real-time performance. [U] / [CO3]

19. Explain how a change in sensor input can change the output of an ECU. [U] / [CO4]

20. Use an embedded audio player to show the main steps involved in playing a stored song. [A] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Describe the basic components of an embedded system and explain how CISC and RISC processors influence system performance [U] / [CO1]

22. Explain event functions in detail and discuss their importance in process coordination and system operation. [U] / [CO3]

23. Explain the operation of a telephone answering machine from receiving an incoming call to playing the recorded message. [U] / [CO4]

[10x2 = 20]